# Banking Intent Classifier — NLP Day 1 Report

## 1. Project Objective

The goal of our project is to classify banking customer messages into 77 intent categories using Natural Language Processing (NLP).

## 2. Dataset Overview

- Dataset: Banking77
- Training examples: 10,003
- Testing examples: 3,080
- Number of intent categories: 77
- Input (X): Customer message text
- Target (Y): Banking intent category

## 3. Data Exploration

We explored the dataset to understand its structure and content.

- We checked the number of categories and examples in each category.
- We read sample messages to understand the meaning of different intents.
- We visualized the number of examples per category.
- We examined the number of words and characters in customer messages.
- We identified the most frequent words in the dataset and in selected categories.

## 4. Data Quality Checks

We checked the dataset for potential quality issues.

- Missing and empty texts or labels.
- Duplicate messages within the training and testing sets.
- Messages shared between the training and testing sets.
- Messages with different labels.
- Very short or very long messages.
- Unexpected characters and possible contact information.

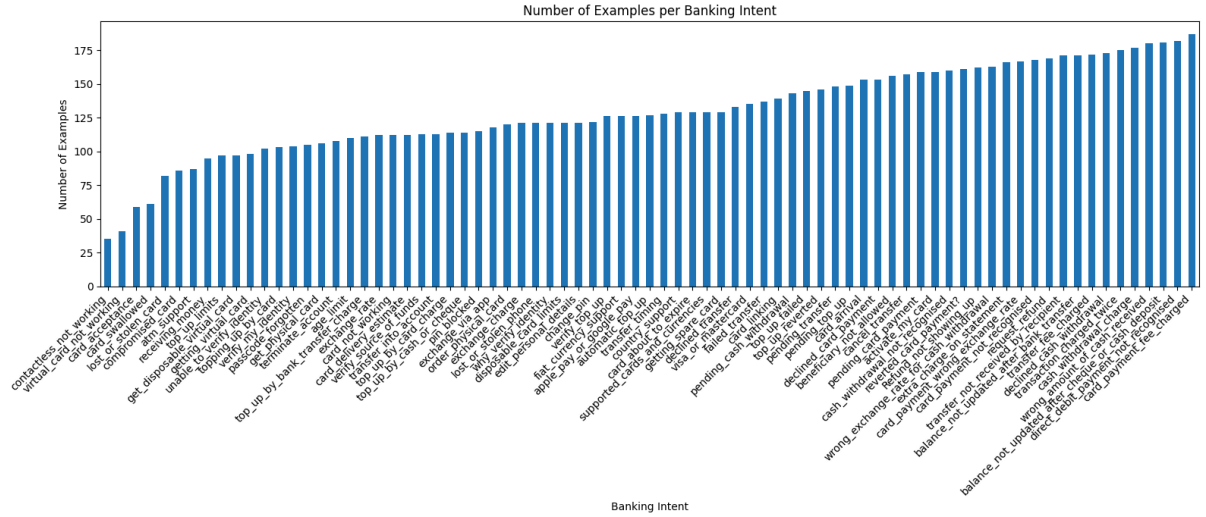
We used these checks to understand the quality of the data. We did not remove examples during exploration.

Results: 0 empty texts, 0 missing labels, 0 duplicates, 0 shared messages between train and test (no data leakage), 0 emails or phone numbers. Only 61 messages with unusual characters and 23 very long messages. The longest message was written by the bank, not a customer.

# 5. Class Distribution

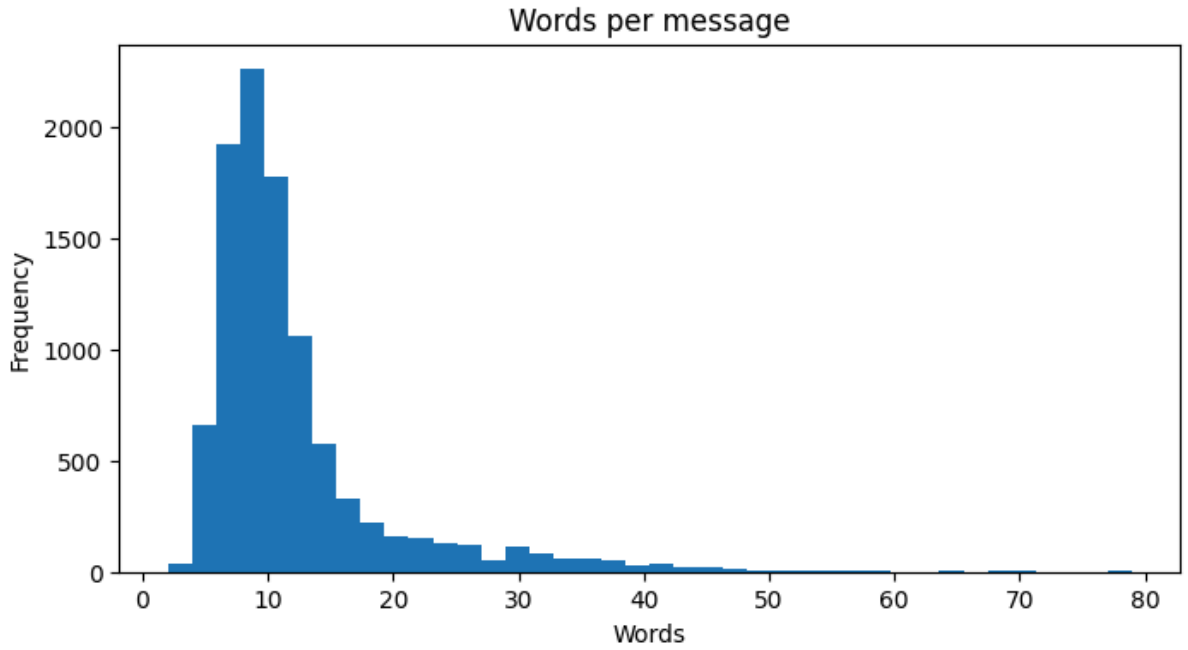
The dataset contains 77 intent categories. We compared the number of examples in different categories to check whether the classes are balanced.

Class distribution is important because some intents may have more examples than others The training data is not balanced: from 35 (contactless_not_working) to 187 (card_payment_fee_charged), about 5 times. The test set is balanced: 40 per intent. The smallest intents are card problems, like lost_or_stolen_card, which is very important for the bank.




# 6. Initial Findings

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- Some banking intents may have similar meanings.
- Message lengths vary between customer queries.
- Common words can help us understand the dataset, but they may not always identify the correct intent.
- Similar intent categories may be difficult for a model to distinguish.
- Data quality and privacy should be considered before modeling.
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- Messages are short, about 10 words (59 characters). Transfer problems are longer (about 17 words). 91% have punctuation. The most common word is "card", but it appears in many intents.




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- Most dangerous mistake: classifying an urgent message, like a stolen card, as a normal question, because the customer could lose money.
- Limitations: small intents have few examples, some intents are very similar (like compromised_card and card_payment_not_recognised), and there is no customer ID to check related messages

## 7. Next Steps

- Review the meanings of all intent categories.
- Discuss data quality findings with the team.
- Decide how to evaluate the model.
- Build a baseline text classification model.
- Compare model performance using suitable evaluation metrics.
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- Our main metric is Macro F1, because the data is not balanced and every intent is equally important. Baseline: TF-IDF + Logistic Regression, then compare with Naive Bayes and Linear SVM.

## 8. Simple Data Dictionary

| Data Type | Description | Column Name |
|---|---|---|
| String (Text) | The customer's banking message used as input (X). | text |
| Categorical (String) | The correct banking intent category that the model predicts (Y). | category |

# Conclusion

Day 1 helped us understand the Banking77 dataset, explore customer messages, and identify potential data quality issues before building a machine learning model.